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Attachments

AI RFI Response FSU 2025

Florida State University is honored to present our insights and recommendations in response to the Request for Information (Published Document: 2025-02305 (90 FR 9088)) concerning the Development of an Artificial Intelligence (AI) Action Plan for the United States. Florida State University commends the Administration's efforts as described in the Presidential Executive Order, dated January 23, 2025, to develop an AI Action Plan to ensure America's global dominance in AI to promote advances for human prosperity, economic success, and national security to strengthen our nation and enhance America's leadership in AI. Drawing upon its extensive expertise and pioneering research in AI and related computer/IT and related technology fields, Florida State University is committed to advancing national AI strategies that will foster innovation, economic growth, and societal benefits.

The perspectives presented represent those of an academic institution with a robust and broad research program across its many colleges, institutes, centers, and programs. Central to its mission is a clear focus on education and training students to become future leaders and important contributors to society. Recognizing that our most meaningful advances are typically made through collaborations, FSU has built an ecosystem that encourages public-private partnerships, and our university's AI strategies reflect this collaborative spirit. Our recommendations for a national AI strategy are presented below.

Overview and Policy Action Framework

The specific recommendations that follow embody a policy framework built on this core principle: *research institutions, such as Florida State University, play an indispensable role in public-private partnerships that advance the country's AI dominance*. Such institutions are key players in **driving technological innovation; training and preparing tomorrow's workforce for impactful, meaningful work; and unleashing the creative and productive potential of AI**. American higher education is unmatched. It is the envy of, and model for, countries around the globe. With innovative public-private partnerships, therefore, our competitive advantage can be parlayed into further American leadership at the cutting edge of AI.

With the tremendous potential that AI affords, policy must be carefully designed and considered to both encourage innovation via incentives while recognizing the government's important role in ensuring the ethical development of AI and respect for citizens' privacy and other cherished freedoms. As the discussion below makes clear, AI stands to change lives as much or more dramatically than many of the groundbreaking technological innovations of the past century. Thus, a holistic view of how AI can impact other important institutions and rights we consider sacred to American democracy is necessary to respect the law of unintended consequences.

To that end, policy that supports fundamental and basic research at research universities is critical. Research universities and their extensive basic and applied research capabilities have been the breeding grounds of important AI innovations in the past and will continue to foster transformative developments going forward. Specifically, university-centered research is critical for the important study of **basic research questions** that might not immediately be valued by the market but will eventually unleash profitable products; by acting as an important check that private innovations provide clear **societal benefits** at acceptable costs; and by providing **mission-critical training** in development, adoption, and utilization of innovations. Without the

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involvement of universities like Florida State, these benefits might be foregone by innovations in the venture-capital-driven private sector.

Substantial funding and support for AI research should be considered for research institutions via traditional methods such as the National Science Foundation (NSF) and the Defense Advanced Research Projects Agency (DARPA). However, in times of competing funding choices, universities are well situated for private-public partnerships for AI innovation. Over the past few decades, universities such as Florida State have focused on relevant research that impacts society throughout the life cycle of product and technology development. Universities have been playing an increasingly vital role in patents and other commercialization activities. Finally, public universities act as the training ground for tomorrow's workforce, so increased hands-on opportunities with the technologies of tomorrow will enable a more competitive and effective workforce. As examples, DARPA grant programs often use public research universities as 'red teams' to serve as reliability and robustness checks on larger, often private, innovation efforts. Further, many NSF programs, such as SBIR, have training and education components that further the impact of grant funding. Such programs should continue to flourish to ensure AI is not only a successful technology but a useful product to drive American economic success.

Specific Recommendations

Recommendation 1: Develop Clear Vision, Objectives, and Frameworks.

The first major step in establishing an AI Action Plan is to define the overarching goals and long-term vision for AI in the US, recognizing the present landscape upon which to build. This involves setting specific, measurable, achievable, relevant, and time-bound (SMART) objectives that align with national interests. The vision should encompass scientific advancement, health and medical discoveries, economic growth, national security, societal well-being, and global harmony. Additionally, a comprehensive regulatory and policy framework is necessary to govern AI development and deployment. This includes creating guidelines for AI use in critical sectors, establishing standards for AI safety and performance, and addressing legal and liability issues. The framework should be flexible to adapt to the rapidly changing AI landscape.

Positioning AI in Human-AI Collaboration – Guardrails and Frameworks

In an era where artificial intelligence (AI) is rapidly transforming various sectors, developing comprehensive guidelines, resources, and toolkits that focus on responsible AI integration is crucial. These resources should include motivating and guiding questions that address where, how, and why AI is used and augmented. By doing so, we can ensure that AI is implemented thoughtfully and ethically, as described below:

- ***Develop open-source frameworks and toolkits – essential to fostering a deeper understanding of AI and its applications.*** These tools can help create communities of practice among educators and students, enabling them to harness AI's power while also recognizing its limitations. We can empower the next generation of AI practitioners by promoting collaboration and knowledge sharing.

- ***Reframe AI as a service.*** By shifting the mindset towards AI and democratizing AI products, we can involve a wider range of users in the development and pilot testing phases. This user-centered approach ensures that AI designs are aligned with the goals and needs of the users, leading to more effective and accessible AI solutions.

- ***Prioritize the development of fundamental science for large AI models and the engineering techniques required to implement them.*** Understanding the mechanisms of AI models is essential for achieving long-term dominance in the field. While the principles of current AI models, such as gradient descent optimization, are well understood, the semantics of the representation space need further exploration. Gaining insights into the structures of the representation space is crucial for many applications.

- ***Develop better algorithms and new model architectures.*** For instance, the published reinforcement learning algorithm in DeepSeek models shows potential, but it does not address many fundamental problems correctly. As demonstrated by the super-human performance of AlphaGo algorithms in the game of Go, correctly implemented reinforcement learning algorithms can lead to significant improvements in various applications.

- ***Share datasets and know-how tricks-of-the-trade to unleash the potential of universities and research institutions.*** By facilitating access to valuable resources and expertise, we can accelerate the pace of AI innovation and discovery.

Recommendation 2: Invest in Research and Development to Ensure Innovation and Competitiveness.

The US should encourage innovation by supporting research and development in AI while also ensuring that regulations do not stifle technological advancements. A strong commitment to R&D is crucial for maintaining innovation in AI. The government should allocate substantial funding to AI research, encourage public-private partnerships, and support academic institutions and AI programs within these institutions. Fostering a vibrant research ecosystem will drive breakthroughs in AI technologies and applications, while cultivating future users and leaders in the field.

<u>AI in Medicine to Promote Enhanced Public Health</u>

AI is emerging as a transformative force in the field of medicine, presenting unprecedented opportunities to enhance patient care, streamline clinical processes, and improve outcomes. As AI technologies continue to advance, their integration into healthcare systems across the US promises to advance medical practice and therefore improve overall health outcomes. Some areas that have seen large potential gains are in enhancing diagnostic accuracy with such tools as AI-powered imaging systems and the use of machine learning algorithms where AI can analyze huge amounts of medical data extremely fast and accurately. Personalized medicine is becoming an increasingly possible through AI with analyses of genetic profiles, lifestyle factors, and other medical history data that can assist physicians in specifically tailoring treatments to

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individual patients. This approach yields much better health outcomes. Beyond improving diagnostics and treatment, AI is extremely useful in streamlining clinical workflows, such as appointment, surgery, and other scheduling, managing patient records, and processing insurance claims. With reduced time spent on administrative burdens, more care can be directed at the patient. Predictive analytics also represents another area where AI is substantially impacting. Patterns in enormous amounts of patient data can be analyzed allowing AI predictions, for example, on the likelihood of certain health conditions or identifying patients or groups of patients at risk for developing diseases. Importantly, this capability enables healthcare providers to shift the focus from intervention to prevention and also provides the opportunity to intervene earlier, likely reducing the incidence and/or severity of diseases. Additionally, AI provides an important tool for educating the public and presenting medical and health information in a variety of platforms, such as bots, chatbots, or other ‘devices’ that can be accessed, as well as AI-powered bots that can provide healthcare/therapy.

These improvements in healthcare, although immense, have challenges and ethical considerations that must be addressed. Patient data and privacy of patient data remain essential, as is the need for total transparency in the AI algorithms. Concerns about potential bias in AI systems could lead to inconsistencies in the delivery of care. These and other concerns strongly suggest the need for the development of robust regulatory frameworks and guidelines essential to ensure that AI is used responsibly and safely in healthcare.

- ***Invest in research and development to advance AI technologies and their applications in medicine and healthcare.***
- ***Provide healthcare professionals with training on AI tools and their integration into clinical practice.***
- ***Establish comprehensive policies to address the ethical and regulatory challenges associated with AI.***
- ***Foster collaboration between healthcare providers, technology companies, and academic institutions to drive innovation.***
- ***Ensure AI applications prioritize patient well-being and personalized care.***

Design for Human-AI Collaboration

The need for explainable AI and human-centered AI research is becoming increasingly important as organizations integrate AI systems alongside human operators. This research aims to enhance our understanding of explainability, transparency, fairness, and interpretability, ensuring that human operators can align with AI's workings. For example, Florida State University researchers have developed the LabGenie system, that leverages Gen AI as a way for patients to better understand medical lab results and ideate on questions they can ask their doctors. This patient aid is an excellent example of how an explainable AI agent can enhance comprehension and empower the public with information, in this case, with their health information.

- ***Develop dynamic AI with situational and contextual awareness to provide personalized experiences in collaborative settings.*** For instance, an AI tool designed to record clinical data and offer best practice guidance must adapt its responses based on individual user needs and contextual awareness. Such tools should cater to various stakeholders, including medical professionals, patients, and community partners, by understanding their unique perspectives and tailoring responses accordingly.

- ***Democratize AI development and usage.*** Generative AI, despite some controversy, democratizes knowledge and empowers individuals by providing access to tools that enhance expression and innovation. For example, generative AI allows non-artists to create art without years of experience, enabling richer content presentation and broader engagement. This can lead to economic growth and support small businesses lacking technical expertise, offering them the necessary infrastructure to start and grow.

- ***Define intellectual property rights for AI-generated content and models to protect creators and innovators.*** The US government should establish clear policies for data collection, storage, and AI model training, aligning with legal and ethical standards.

- ***Additionally, develop frameworks for monetizing AI, such as licensing and subscription models, to enable fair compensation and scalable growth in the AI industry.*** Academia and higher educational institutions are major consumers of AI-enabled products (such as GPT models and extensions for varied workflows in text processing and image analysis). Therefore, having custom licenses and academic rates will bolster AI innovation and partnerships between academia and industry.

- ***Enhance profitability using AI as a tool for organizations.*** Essential is an understanding of industry's AI adoption and utilization strategies. Improved AI learning and training methods should address inherent incentive issues in AI training, unlocking the potential for real-time personalization and learning within AI mechanisms. Understanding how self-interested humans verify AI training data or use its recommendations is critical for developing the next generation of AI models.

- ***Ensure a heterogeneous infrastructure to make AI reliable in safety-critical areas like manufacturing.*** Policies should focus on researching and developing ultra-reliable infrastructure to integrate AI and applications. High-performance computing (HPC) provides the computational power needed to train and run complex AI models, enabling the processing of massive datasets and complex algorithms. Heterogeneous computing, utilizing different processing units optimized for specific tasks, enhances AI computations' reliability, efficiency, and speed. Together, HPC and heterogeneous computing accelerate AI research and development by providing the necessary infrastructure for advanced AI models.

Integrating various processors, such as GPUs, TPUs, FPGAs, and CPUs, tailored for different computing environments (on devices or in the cloud), allows for optimized processing of specific AI tasks, leading to faster training times and improved performance. Both HPC and heterogeneous computing architectures can scale to meet AI research's growing computational demands, enabling increasingly complex AI applications.

- ***Incentivize research that supports long-term, heterogeneous computing systems for integrating AI applications.*** Public universities, like Florida State University, with robust computing infrastructures, play a crucial role in maintaining AI dominance. Therefore, new policies should support and incentivize research in such institutions.

Recommendation 3: Enhance Data Infrastructure and Access.

Data is the lifeblood of AI. Ensuring that data is accessible, high-quality, and secure is fundamental to AI development. The government should create policies that facilitate data sharing while respecting and ensuring privacy and mitigating security concerns.

<u>Frameworks and Expansions of Data Storage Infrastructure</u>

Data is the cornerstone of AI development, influencing model performance, fairness, and reliability. To foster innovation while safeguarding privacy and security, a structured approach to data infrastructure is essential. This approach should incorporate data injection, data provenance, data governance, and controlled data access.

- ***Adopt a federated architecture within a national AI data infrastructure to support secure data sharing across institutions while maintaining data privacy and security.*** Federated learning and differential privacy techniques can be leveraged to ensure privacy-preserving data access while enabling AI models to learn from distributed datasets. Establishing federated AI data hubs for critical domains such as healthcare, climate science, and cybersecurity is crucial. Additionally, AI models trained on distributed datasets should adhere to Fair Information Practices (FIP), and secure sandbox environments should be provided for AI researchers to test models on real-world data without direct access.

- ***Ensure trust in AI systems through the ability to trace the origins and modifications of data.*** Embedding data provenance mechanisms within the infrastructure ensures transparency, reproducibility, and accountability. Mandating machine-readable metadata standards like the FAIR principles (Findability, Accessibility, Interoperability, and Reusability) is essential. Implementing blockchain-based audit trails and developing explainable AI (XAI) tools help users understand how data impacts AI decision-making.

- ***Balance data access with privacy preservation, which is crucial for ethical AI development.*** AI governance frameworks should enforce tiered access models based

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on user credentials, research purposes, and ethical review processes. Establishing data access policies that align with regulations, promoting synthetic data and secure multi-party computation (SMPC) for AI model training, and developing an AI Data Commons for secure access to high-quality public-sector datasets are necessary actions.

- ***Make national investments to curate, standardize, and expand datasets; create data annotation frameworks; develop public-private partnerships; and incentivize open-source AI datasets while enforcing responsible data-sharing practices.*** Public universities can enable and engage communities for tight integrations, ensuring AI benefits small companies and essential community members.

Recommendation 4: Create Public-Private and Other Partnerships.

Working together to develop standards and practices for AI, promoting cooperation, and addressing national and global challenges will undoubtedly ensure greatest achievements and most efficiencies. Collaboration between the government, industry, academia, and non-profit organizations is vital for the progress and effective implementation of AI. Establishing partnerships and collaborative frameworks will enable knowledge exchange, resource sharing, and coordinated efforts to address AI challenges and opportunities.

- ***Build incentive structures within private contracting for collaboration with public research universities and ensure requirements for independent testing and validation of products.***
- ***Require educational training and access component to private contract to allow for training of the next generation of AI users and developers.***
- ***Foster interdisciplinary collaborations across computing, social sciences and humanities to develop a roadmap for smart futures.*** This could also entail establishment of collaborative partnerships/programs, scholarly exchanges and innovation centers that connect academia, industry and government to break silos of expertise and create a bridge to connect ideas, developing a shared purpose towards future visions and goals.
- ***Create more platforms for sharing academic insights and facilitating industry-academic partnerships.*** For example, Florida State University has partnered with Microsoft to ideate and develop novel, custom tools that can help educators build on Gen AI powered agents to create instructional content and enrich classroom learning.
- ***Engage with industry in student and faculty discussion forums such as conferences, campus-wide events and consortia.*** For example, Florida State University has several centers and collaboration catalysts (such as its Innovation Hub), where industry partners can present and discuss research-industry partnership opportunities to better engage with students, faculty, and the public.

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Recommendation 5: Ensure National Security.

Ensuring the security of AI systems is paramount. AI-driven national security strategies focus on the identification of potential *problems*, *risks*, and *challenges* arising from AI advancements. Rather than focusing on the broad issues posed by AI, a concerted focus should be on the most critical problems immediately confronting us. Four potential problems related to national security are identified and elaborated upon below.

<u>Deepfakes and Fabrication</u>
Florida State University research highlights the growing misuse of deepfakes for fraud, identity theft, and misinformation, threatening national security and public trust. AI-generated deepfakes can impersonate authority figures, bypass facial recognition, and manipulate individuals.

- ***Combat deepfakes through robust detection methods.*** Examples of such methods include AI-powered verification tools, real-time detection, and regulatory measures such as watermarking.

- ***Build public AI literacy through education and awareness campaigns.*** These initiatives can help individuals recognize and resist digital manipulation.

- ***Integrate technological, regulatory, and educational initiatives to enhance cybersecurity, prevent fraud, and safeguard public trust in digital content.***

<u>AI-powered Deliberate Disinformation Campaigns</u>
Research at Florida State University highlights that AI-powered deliberate disinformation campaigns pose a significant threat, especially when exploited by malicious actors or adversarial nations. Preventing such misuse is critical to maintaining societal trust and security. By developing resilient AI systems, we can:

- ***Enhance our defense against cyber threats, including misinformation campaigns that distort reality, by developing resilient AI systems.*** However, biased or misused AI algorithms can amplify these risks. Ensuring transparency and explainability in AI decisions is vital to foster accountability and trust.

- ***Leverage AI to detect and prevent misinformation, identify foreign election interference, and develop voter fraud detection tools to protect the integrity of democratic processes.***

<u>Lone Wolf Terrorists and Rogue Nation States</u>
FSU research can inform the development of AI-powered tools to detect cyber insider threats, lone wolf terrorists, and rogue nation-states by analyzing interpersonal and group communication. Identifying these threats is critical for national security, as cyberspace has no borders but can be safeguarded using advanced deception detection technologies. While traditional polygraph solutions rely on physiological data, AI-driven communication analysis can infer deception and identify malicious intent in digital environment.

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- ***Strengthen national security through further research to enhance AI-driven cyber behavioral analysis, sentiment detection, and intent assessment.*** These technologies can detect emerging threats, assess online actors' trustworthiness, and prevent attacks.
- ***Integrate AI-powered solutions into national security frameworks will improve early threat detection and response, ultimately protecting individuals, communities, and the nation.***

<u>Traceability of Anonymity Online</u>
Florida State University research can inform strategies to enhance the traceability of online anonymity, a critical issue as large-scale AI initiatives like the Stargate Project that expands digital infrastructure in the United States. While AI-driven platforms offer significant advancement, they also introduce security risks by enabling anonymous bad actors to evade detection. Anonymity can obscure accountability, facilitating cybercrime, long-wolf terrorism, and state-sponsored threats. Additionally, AI-generated false data and the complexity of network traffic make it difficult to trace digital identities, complicating law enforcement efforts.

- ***Develop AI-powered solutions for identifying and tracking malicious online activity while balancing privacy and security concerns to mitigate these risks.*** Strengthening digital forensics, enhancing VoIP traceability, and leveraging AI for real-time anomaly detection will be critical in safeguarding national security and ensuring the integrity of digital ecosystems.

- ***Develop AI platforms that are more trustworthy and reliable than the competition.***

- ***Create systems that enhance imperfect and biased human decision-making in both commercial and military settings.*** Trust will be essential for widespread public adoption of AI technology and for effective collaboration between human and robotic troops in military operations.

Recommendation 6: Protect Data Privacy and Security.

Speaking broadly, there need to be coalitions across the government, private and public sectors to protect people's privacy rights, and to make good decisions concerning how AI is applied and used in surveillance and monitoring. How various priorities and rules would interact with existing privacy laws requires additional assessment. Additionally, infrastructure and resources for seeking redressal in the case of violation of legally protected privacy needs to be created.

For its own part, the government should implement measures to protect AI systems from threats, including cyber-attacks, exploitation, and misuse. Data privacy and robust security measures must be in place to protect individuals' personal information and to prevent data breaches. Five potential vulnerabilities related to national security are identified and elaborated upon below that need to be addressed with top priority.

Cyber-Physical Critical Infrastructure

Our research demonstrates that cyber-physical critical infrastructure is increasingly vulnerable as generative AI enables the creation of sophisticated code for programmable logic controllers (PLCs), which are vital for industrial systems. Hackers can exploit AI-generated code to infiltrate and manipulate vital infrastructure, including water treatment plants, electrical grids, and transportation networks, leading to potential equipment failure or public safety risks. Safeguarding these critical infrastructures against cyber-physical threats is a growing priority.

- ***Deploy AI-enhanced sensors to detect early failure signs in critical systems and combine AI predictions with human expertise to avoid false positives.***

AI-Powered Cyber-Physical Attacks

Our work demonstrates that AI-powered cyber-physical attacks, particularly those leveraging adversarial machine learning, have greatly increased the sophistication of cybersecurity threats. Polymorphic malware, such as the Black Mamba ransomware, highlights how AI can predict and bypass traditional defense mechanisms, rendering current cybersecurity technologies ineffective. Hackers exploit AI techniques to evade endpoint detection and response by dynamically retrieving synthesized and polymorphic code from trusted AI sources, allowing malware to carry out harmful actions without triggering security alerts. Furthermore, generative AI complicates security by creating unique malware payloads, making it increasingly challenging to distinguish between benign and malicious code, ultimately exposing systems to undetectable threats.

- ***Invest in cyber-security training programs and the development of cyber-security platforms to combat cyber-physical attacks.***

Malicious GPTs and AI Exploitation

Florida State University research highlights that hackers can exploit malicious and generative pre-trained transformers (GPTs) to undermine the accuracy and reliability of AI/ML systems through various attack strategies. These include poisoning attacks, where fake data is injected into training datasets, evasion attacks that manipulate input data, and model tampering, which alters AI model parameters. These tactics collectively weaken AI-driven decision-making, making cybersecurity defenses more vulnerable to manipulation.

- ***Implement security-by-design principles, conducting adversarial testing, establishing AI security initiatives, and enhancing AI interpretability for transparency and accountability.*** Additionally, AI has significantly improved the effectiveness of Distributed Denial-of-Service (DDoS) attacks and malware performance, enabling attackers to execute more powerful, adaptive, and persistent cyber threats and disruptions.

AI Manipulation

Florida State University institutional research demonstrates that AI manipulation has become a significant concern, as AI-generated output can influence/deceive people's perception and decision-making. One notable example is AI-generated phishing attacks, where AI creates

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highly convincing fake emails or websites that are difficult to distinguish from legitimate ones. Additionally, AI can craft personalized messages, increasing the likelihood of successful deception and potentially leading to data theft, influencing mass perceptions, and other malicious activities.

- ***Direct research funding toward AI-generated content detection tools, fostering collaboration between cybersecurity experts, AI researchers, and policymakers to stay ahead of evolving cybercriminal tactics, and establishing stronger safeguard and legislation against AI-based fraud.***

Recommendation 7: Promote Ethical AI Development.

Establishment of guidelines that evolve with the further development of AI is essential to ensure AI systems are designed and used ethically and avoid biases. AI development and deployment must adhere to ethical standards that ensure fairness, accountability, and transparency.

- ***Establish a thoughtful set of ethical guidelines to prevent biases, protect privacy, and promote the responsible and ethical use of AI technologies.*** This should include input from diverse stakeholders, including ethicists, technologists, and affected communities.

<u>Human-in-the-loop Configuration of Human-AI Interaction</u>

A clear need exists to understand the intricacies of human experience and how they shape the way AI is perceived, adopted and deployed. There needs to be a focus on human-in-the-loop frameworks for designing and developing AI technologies, so that stakeholder voices are valued and embedded in the AI usage pipeline.

- ***Repurpose the focus of AI innovation from only technical to sociotechnical.*** Increasingly, AI artifacts are finding use and extensive deployment in varied societal contexts, increasing the need to raise awareness and consciousness towards how AI is designed and developed through a human-centered approach.

<u>AI Bias</u>

Research at Florida State University highlights that AI bias is a significant concern, as it can lead to discriminatory outcomes stemming from skewed or non-representative datasets. Just as human's biases are influenced by personal preferences, AI systems can inherit similar biases, which cannot be entirely eliminated. As AI systems are trained on large volumes of human data, privacy risks also increase. While federated learning techniques offer potential for learning from distributed data, they also pose challenges in ensuring unbiased data correlation.

- ***Implement predevelopment bias assessments for AI systems.***

- ***Incorporate bias detection and auditing into AI models.***

- ***Ensure AI systems use broad, representative datasets that reflect population demographics.***

- ***Develop human-in-the-loop systems for sensitive decisions.***

- ***Establish nonpartisan oversight for election security tools.***

Recommendation 8: Invest in Workforce Development.

Investing in education and training programs is essential for preparing the workforce to handle and optimally apply the changes brought by AI. Building a skilled workforce is essential for sustaining AI growth.

- ***Invest in education and training programs at all levels, from K-12 to higher education and lifelong learning.*** Emphasizing STEM (Science, Technology, Engineering, and Mathematics) education, interdisciplinary studies, and AI-specific curricula will prepare the next generation of AI professionals.

- ***Integrate AI-specific curricula in all post-graduate, professional educational programs to ensure that AI tools are being used effectively in all professions.***

<u>Curriculum Development to Embrace and Collaborate with AI</u>

Human-AI collaboration will require transforming the future AI workforce's preparedness, emphasizing protection against overreliance on AI. This involves balancing technical and societal considerations, fostering interdisciplinary training, and developing novel pedagogical approaches. Literacy initiatives drawing on humanities and social sciences are crucial for cultivating ethical and moral consciousness towards responsible AI use.

- ***Create experiential educational pathways and AI-assisted learning tools to expand capacities and enable non-programmers to build systems.***

- ***Make AI products accessible and provide essential guidelines for academic use.***

- ***Ensure safety training and curricula to enhance critical consciousness and situational awareness, especially in high-risk conditions like armed forces operations.***

The following sections illustrate how AI can be applied effectively in teaching, training, and learning efforts critical to a much more robust development and use of AI's full potential.

<u>Teacher Training and AI Innovation</u>

Florida State University has led several professional development sessions associated with its INSPIRE Initiative to train K-12 educators to have a deeper understanding of AI and associated areas (such as generative AI, prompt engineering, data science and responsible AI).

- ***Support teacher training initiatives to strengthen the capabilities of educators to better equip and prepare the future AI workforce.*** Such initiatives can help create and expand existing teacher certifications and professional development goals that are essential for evolving teaching needs in STEM education as AI and data science skills become essential for the future workforce.

K–12 AI Literacy Programs

Foundational skills are essential for creating an AI-ready workforce pipeline.

- ***Embed AI literacy within K-12 curricula, because early exposure to AI concepts develops critical evaluation skills that prepare students to identify algorithmic biases and understand ethical implications of AI technologies.*** These foundational skills are essential for creating an AI-ready workforce pipeline. Students exposed to these programs transition from passive AI consumers to informed contributors within the AI ecosystem.

Advanced Self-Evaluation and Skills Assessment

Florida State University's "Khan Academy on Steroids" platforms provide sophisticated self-evaluation tools that offer granular insights into learning deficiencies and strengths. These AI-driven diagnostic tools generate real-time assessments with personalized recommendations, fostering self-regulated learning and autonomous intellectual growth - critical skills for workforce adaptability in an AI-driven economy.

AI-Driven Career Navigation

Florida State University's research on predictive analytics for career development demonstrates how AI can assess labor market trends and align professional development with emerging industry demands. Machine learning algorithms identify skill gaps and recommend tailored training modules, creating dynamic career roadmaps that help workers adapt to technological disruption and maintain employability.

Immersive Training Environments

Florida State University's work with AI-powered VR training modules revolutionizes vocational education by offering immersive, experiential learning environments. These AI-driven simulations model complex job-specific scenarios and dynamically adjust difficulty based on learner performance, providing risk-free skill acquisition particularly valuable in high-stakes industries such as healthcare, manufacturing, and cybersecurity.

Higher Education Research Integration

Florida State University's integration of AI in higher education shows how AI-powered research assistants streamline scholarly workflows, while AI-driven tutoring platforms provide targeted instructional support in STEM fields. AI-facilitated peer collaboration platforms foster interdisciplinary research synergies essential for breakthrough innovation in AI development.

Personalized Learning Pathways

Our research demonstrates that AI-powered adaptive learning systems significantly enhance educational outcomes by dynamically analyzing student performance and providing targeted interventions. These systems scaffold instruction based on individual progress trajectories, ensuring differentiated pedagogical support that accelerates skill acquisition - a model that should be expanded nationwide through strategic federal investment.

<u>Adult Basic Education Enhancement</u>
Florida State University’s work with Adult Basic Education (ABE) programs showcases how AI can democratize access to high-quality education for working adults. AI-driven language models facilitate tailored reading comprehension passages and contextualized skill-building exercises, while adaptive platforms adjust content complexity based on learner progression. This approach is particularly effective for adults balancing employment and family obligations, who require flexible, personalized educational modalities. Each area should consider providing examples that address this recommendation to provide context for our perspectives on policy implications.

Recommendation 9: Promote International Cooperation.

AI development is a global endeavor. The US should engage in international cooperation to harmonize standards, share best practices, and address global challenges. Participating in international forums and collaborating with other nations will strengthen the global AI ecosystem and ensure that AI benefits are widely shared. International standards and best practices for AI must be developed and promoted. Global challenges can be addressed in collaboration using the power of AI.

<u>Exchange of Ideas, Innovation, and Cooperation</u>

- ***Create strategic alliances for innovating and piloting AI usage and deployment.*** Understanding policy and future growth in the AI sector within the global economy helps us grasp how different nations may approach AI innovation and preparedness in the coming years.

- ***Develop and cultivate crucial research partnerships.*** For example, French institutions and private companies have been key players in developing humanoid robots, while Japan has long been at the forefront of humanoid robotics. It is vital to understand the momentum, market, and ways to build partnerships with these global giants, along with their business models and innovative visions.

- ***As data becomes increasingly valuable to AI, protect it just like any natural resource.*** Currently, there are relatively few guardrails to protect individuals' privacy and safety in this new digital world. While sharing the digital commons across the world is a hallmark of the 21st century, ***we must recognize the new landscape and proactively work to protect citizens while maintaining essentially 'free trade' in data.***

- ***Increase international cooperation for data security and protection from manipulation and fraud.*** The international protection of intellectual property rights, such as copyrights, patents, and trademarks, has been a significant issue in the past, and developments in AI make these threats even more real.

Conclusion

Maintaining the US's position as the global leader in AI is important for national advancement, productivity, protection, and prosperity, and to ensure that AI serves human flourishing. Universities provide an essential engine that drives technological innovation through leading multidisciplinary research programs and by training the next generation of AI developers and users, as well as producing advances in related fields necessary to fuel further evolution and innovation. We, however, recognize that this effort cannot be accomplished by a single group alone and must be a concerted collaboration between government, public, and private partners, all working together toward shared goals shaped by federal policies.

We value this opportunity to provide feedback and guidance as you move forward in setting policies and establishing the necessary infrastructure that will enable the US to lead in this monumental opportunity to harness the power and potential of AI. We hope to contribute in a meaningful way in helping America stand firm as a global leader in this area.

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